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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Kinetic Theory of Gases

JEE Main Physics · Previous Year Questions with Answer Key

167	152	2002 – 2026	English
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MCQ (Single Correct Answer)	151 questions
Numerical	16 questions
Answer key	at the end

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MCQ (Single Correct Answer) 151 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

If 2 mole of an ideal monoatomic gas at temperature T, is mixed with 6 mole of another ideal monoatomic gas at temperature $2T$ then the temperature of mixture is:

(A) $\frac{5}{2}T$ (B) $\frac{5}{4}T$ (C) $\frac{7}{2}T$ (D) $\frac{7}{4}T$

Open solution

Q2. JEE Main 2026 (Online) 6th April Morning Shift

Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K . Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K . The final pressure in the vessels is ____ kPa .

(A) 100 (B) 120 (C) 90 (D) 105

Open solution

Q3. JEE Main 2026 (Online) 5th April Morning Shift

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Statement I : Change in internal energy of a system containing n mole of ideal gas can be written as $\Delta \mathrm{U}=n \mathrm{C}_v\left(T_{\mathrm{f}}-T_i\right)=\frac{nR}{\gamma-1}\left(T_{\mathrm{f}}-T_i\right)$, where $\gamma=\frac{C_p}{C_v}, T_i=$ initial temperature, $T_{\mathrm{f}}=$ final temperature.

Statement II : Relation between degree of freedom f and $\gamma\left(=C_p / C_v\right)$ is $\left(\gamma=1+\frac{2}{f}\right)$

Choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is true but $\mathbf{R}$ is false

(D) A is false but $\mathbf{R}$ is true

Open solution

Q4. JEE Main 2026 (Online) 4th April Evening Shift

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : If the average kinetic energy of H_2 and O_2 molecules, kept in two different sized containers are same, then their temperatures will be same.

Reason R : The r.m.s speed of H_2 and O_2 molecules are same at same temperature.

Choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is true but $\mathbf{R}$ is false

(D) A is false but $\mathbf{R}$ is true

Open solution

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Q5. JEE Main 2026 (Online) 4th April Morning Shift

One gas of n_1 mole of molecules at temperature T_1, volume V_1, and pressure P_1, and another gas of n_2 mole of molecules at temperature T_2, volume V_2, and pressure P_2, are mixed resulting in pressure P and volume V of the mixture. The temperature of the mixture is ____ .

(A) $\frac{(T_1+T_2)}{2}$

(B) $\frac{T_1T_2PV}{T_2P_1V_1+T_1P_2V_2}$

(C) $\frac{(T_2P_1V_1+T_1P_2V_2)}{T_1T_2PV}$

(D) $\frac{[T_1-T_2]}{2}$

Open solution

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

A mixture of carbon dioxide and oxygen has volume 8310 cm^3, temperature 300 K, pressure 100 kPa and mass 13.2 g. The number of moles of carbon dioxide and oxygen gases in the mixture respectively are _____.

(Assume both carbon dioxide and oxygen gases behave like ideal gases) [R = 8.31 J/mol K]

(A) 0.15 and 0.18 (B) 0.25 and 0.08 (C) 0.21 and 0.12 (D) 0.13 and 0.20

Open solution

Q7. JEE Main 2026 (Online) 28th January Evening Shift

The mean free path of a molecule of diameter 5×10^{-10} m at the temperature 41°C and pressure 1.38×10^5 Pa, is given as _____ m. (Given $k_B = 1.38 \times 10^{-23}$ J/K).

(A) $2\sqrt{2} \times 10^{-10}$ (B) $10\sqrt{2} \times 10^{-8}$ (C) $2\sqrt{2} \times 10^{-8}$ (D) 2×10^{-8}

Open solution

Q8. JEE Main 2026 (Online) 22nd January Evening Shift

Consider two boxes containing ideal gases A and B such that their temperatures, pressures and number densities are same. The molecular size of A is half of that of B and mass of molecule A is four times that of B. If the collision frequency in gas B is 32×10^{18}/s then collision frequency in gas A is ____ /s.

(A) 8×10^8 (B) 2×10^8 (C) 32×10^8 (D) 4×10^8

Open solution

Q9. JEE Main 2026 (Online) 21st January Evening Shift

The r.m.s. speed of oxygen molecules at 47 °C is equal to that of the hydrogen molecules kept at _____ °C. (Mass of oxygen molecule/mass of hydrogen molecule = 32/2)

(A) -100 (B) -253 (C) -20 (D) -235

Open solution

Q10. JEE Main 2025 (Online) 7th April Evening Shift

The helium and argon are put in the flask at the same room temperature (300 K). The ratio of average kinetic energies (per molecule) of helium and argon is:

(Give: Molar mass of helium = 4 g/mol, Molar mass of argon = 40 g/mol)

(A) $1 : \sqrt{10}$ (B) 10 : 1 (C) 1 : 10 (D) 1 : 1

Open solution

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Q11. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Match the List I with List II

	List - I		List - II
(A)	Triatomic rigid gas	(I)	$\frac{C_p}{C_v}=\frac{5}{3}$
(B)	Diatomic non-rigid gas	(II)	$\frac{C_p}{C_v}=\frac{7}{5}$
(C)	Monoatomic gas	(III)	$\frac{C_p}{C_v}=\frac{4}{3}$
(D)	Diatomic rigid gas	(IV)	$\frac{C_p}{C_v}=\frac{9}{7}$

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II

(B) A-II, B-IV, C-I, D-III

(C) A-IV, B-II, C-III, D-I

(D) A-III, B-II, C-IV, D-I

Q12. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

There are two vessels filled with an ideal gas where volume of one is double the volume of other. The large vessel contains the gas at 8 kPa at 1000 K while the smaller vessel contains the gas at 7 kPa at 500 K . If the vessels are connected to each other by a thin tube allowing the gas to flow and the temperature of both vessels is maintained at 600 K , at steady state the pressure in the vessels will be (in kPa).

(A) 24

(B) 4.4

(C) 18

(D) 6

Q13. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

The mean free path and the average speed of oxygen molecules at 300 K and 1 atm are $3\times10^{-7}\,\mathrm{m}$ and $600\,\mathrm{m/s}$, respectively. Find the frequency of its collisions.

(A) $5\times10^{8}/\mathrm{s}$

(B) $9\times10^{5}/\mathrm{s}$

(C) $2\times10^{10}/\mathrm{s}$

(D) $2\times10^{9}/\mathrm{s}$

Q14. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Pressure of an ideal gas, contained in a closed vessel, is increased by 0.4% when heated by $1^{\circ}\mathrm{C}$. Its initial temperature must be:

(A) 2500 K

(B) $25^{\circ}\mathrm{C}$

(C) $250^{\circ}\mathrm{C}$

(D) 250 K

Q15. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

The ratio of vapour densities of two gases at the same temperature is $\frac{4}{25}$, then the ratio of r.m.s. velocities will be :

(A) $\frac{5}{2}$

(B) $\frac{25}{4}$

(C) $\frac{4}{25}$

(D) $\frac{2}{5}$

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Q16. JEE Main 2025 (Online) 28th January Evening Shift

The kinetic energy of translation of the molecules in 50 g of CO_2 gas at 17°C is :

(A) 4205.5 J (B) 3582.7 J (C) 3986.3 J (D) 4102.8 J

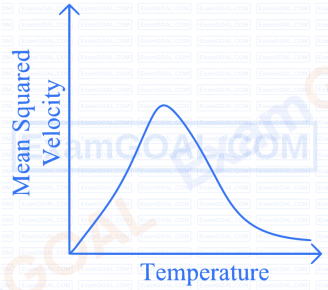
Open solution

Q17. JEE Main 2025 (Online) 28th January Morning Shift

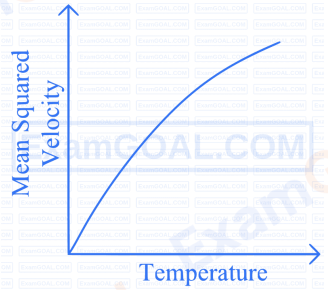
For a particular ideal gas which of the following graphs represents the variation of mean squared velocity of the gas molecules with temperature?

Open solution

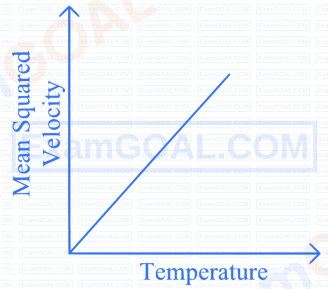
(A)



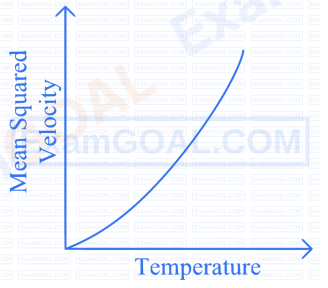
(B)



(C)



(D)



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Q18. JEE Main 2025 (Online) 22nd January Evening Shift

For a diatomic gas, if $\gamma_1 = \left(\frac{Cp}{Cv}\right)$ for rigid molecules and $\gamma_2 = \left(\frac{Cp}{Cv}\right)$ for another diatomic molecules, but also having vibrational modes. Then, which one of the following options is correct? (Cp and Cv are specific heats of the gas at constant pressure and volume)

(A) $\gamma_2 < \gamma_1$ (B) $\gamma_2 > \gamma_1$ (C) $\gamma_2 = \gamma_1$ (D) $2\gamma_2 = \gamma_1$

Open solution

Q19. JEE Main 2024 (Online) 9th April Evening Shift

The temperature of a gas is $-78^{\circ}\mathrm{C}$ and the average translational kinetic energy of its molecules is K. The temperature at which the average translational kinetic energy of the molecules of the same gas becomes $2\,\mathrm{K}$ is :

(A) $-78^{\circ}\mathrm{C}$ (B) $127^{\circ}\mathrm{C}$ (C) $-39^{\circ}\mathrm{C}$ (D) $117^{\circ}\mathrm{C}$

Open solution

Q20. JEE Main 2024 (Online) 8th April Evening Shift

Given below are two statements :

Statement (I) : The mean free path of gas molecules is inversely proportional to square of molecular diameter.

Statement (II) : Average kinetic energy of gas molecules is directly proportional to absolute temperature of gas.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Open solution

Q21. JEE Main 2024 (Online) 8th April Morning Shift

A mixture of one mole of monoatomic gas and one mole of a diatomic gas (rigid) are kept at room temperature $(27^{\circ}\mathrm{C})$. The ratio of specific heat of gases at constant volume respectively is:

(A) $\frac{3}{2}$ (B) $\frac{3}{5}$ (C) $\frac{7}{5}$ (D) $\frac{5}{3}$

Open solution

Q22. JEE Main 2024 (Online) 6th April Evening Shift

Energy of 10 non rigid diatomic molecules at temperature T is :

(A) 35 RT (B) $\frac{7}{2}$ RT (C) 70 K_BT (D) 35 K_BT

Open solution

Q23. JEE Main 2024 (Online) 6th April Morning Shift

A sample contains mixture of helium and oxygen gas. The ratio of root mean square speed of helium and oxygen in the sample, is :

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{4}$ (C) $\frac{2\sqrt{2}}{1}$ (D) $\frac{1}{32}$

Open solution

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Q24. JEE Main 2024 (Online) 5th April Evening Shift

If n is the number density and d is the diameter of the molecule, then the average distance covered by a molecule between two successive collisions (i.e. mean free path) is represented by :

(A) $\frac{1}{\sqrt{2}\mathrm{n}\pi\mathrm{d}^2}$ (B) $\frac{1}{\sqrt{2}n^2\pi^2d^2}$

(C) $\frac{1}{\sqrt{2n\pi d^2}}$ (D) $\sqrt{2}\mathrm{n}\pi\mathrm{d}^2$

Open solution

Q25. JEE Main 2024 (Online) 5th April Morning Shift

If the collision frequency of hydrogen molecules in a closed chamber at $27^{\circ}\mathrm{C}$ is Z, then the collision frequency of the same system at $127^{\circ}\mathrm{C}$ is :

(A) $\frac{\sqrt{3}}{2}\mathrm{Z}$ (B) $\frac{2}{\sqrt{3}}\mathrm{Z}$ (C) $\frac{3}{4}\mathrm{Z}$ (D) $\frac{4}{3}\mathrm{Z}$

Open solution

Q26. JEE Main 2024 (Online) 4th April Evening Shift

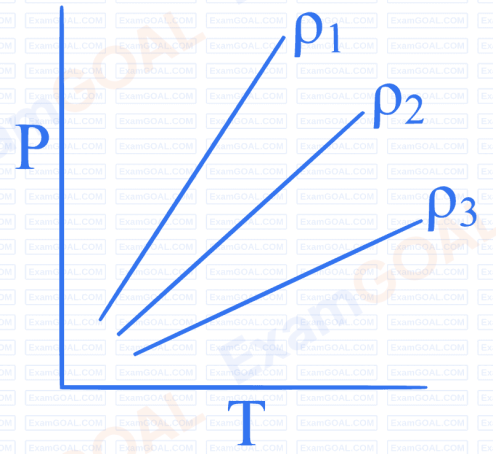
The translational degrees of freedom (f_t) and rotational degrees of freedom (f_r) of $\mathrm{CH_4}$ molecule are:

(A) $f_t = 2$ and $f_r = 2$ (B) $f_t = 3$ and $f_r = 3$ (C) $f_t = 3$ and $f_r = 2$ (D) $f_t = 2$ and $f_r = 3$

Open solution

Q27. JEE Main 2024 (Online) 4th April Morning Shift

P-T diagram of an ideal gas having three different densities ρ_1, ρ_2, ρ_3 (in three different cases) is shown in the figure. Which of the following is correct :



(A) $\rho_1 > \rho_2$ (B) $\rho_2 < \rho_3$ (C) $\rho_1 = \rho_2 = \rho_3$ (D) $\rho_1 < \rho_2$

Open solution

Q28. JEE Main 2024 (Online) 1st February Evening Shift

If the root mean square velocity of hydrogen molecule at a given temperature and pressure is $2\ \mathrm{km/s}$, the root mean square velocity of oxygen at the same condition in $\mathrm{km/s}$ is :

(A) 1.0 (B) 1.5 (C) 2.0 (D) 0.5

Open solution

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Q29. JEE Main 2024 (Online) 1st February Morning Shift

Two moles of a monoatomic gas is mixed with six moles of a diatomic gas. The molar specific heat of the mixture at constant volume is :

(A) $\frac{3}{2}\mathrm{R}$ (B) $\frac{7}{4}\mathrm{R}$ (C) $\frac{5}{2}\mathrm{R}$ (D) $\frac{9}{4}\mathrm{R}$

Open solution

Q30. JEE Main 2024 (Online) 31st January Evening Shift

A gas mixture consists of 8 moles of argon and 6 moles of oxygen at temperature T. Neglecting all vibrational modes, the total internal energy of the system is:

(A) 29 RT (B) 27 RT (C) 20 RT (D) 21 RT

Open solution

Q31. JEE Main 2024 (Online) 31st January Morning Shift

The parameter that remains the same for molecules of all gases at a given temperature is :

(A) kinetic energy (B) mass (C) momentum (D) speed

Open solution

Q32. JEE Main 2024 (Online) 30th January Evening Shift

If three moles of monoatomic gas $\left(\gamma=\frac{5}{3}\right)$ is mixed with two moles of a diatomic gas $\left(\gamma=\frac{7}{5}\right)$, the value of adiabatic exponent γ for the mixture is

(A) 1.35 (B) 1.52 (C) 1.40 (D) 1.75

Open solution

Q33. JEE Main 2024 (Online) 30th January Morning Shift

At which temperature is the r.m.s. velocity of a hydrogen molecule equal to that of an oxygen molecule at $47^{\circ}\mathrm{C}$?

(A) 20 K (B) 80 K (C) 4 K (D) -73 K

Open solution

Q34. JEE Main 2024 (Online) 29th January Evening Shift

The temperature of a gas having 2.0×10^{25} molecules per cubic meter at $1.38\ \mathrm{atm}$ (Given, $\mathrm{k}=1.38 \times 10^{-23} \mathrm{JK}^{-1}$) is :

(A) 500 K (B) 300 K (C) 200 K (D) 100 K

Open solution

Q35. JEE Main 2024 (Online) 29th January Evening Shift

N moles of a polyatomic gas $(f=6)$ must be mixed with two moles of a monoatomic gas so that the mixture behaves as a diatomic gas. The value of N is :

(A) 6 (B) 2 (C) 4 (D) 3

Open solution

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Q36. JEE Main 2024 (Online) 29th January Morning Shift

Two vessels A and B are of the same size and are at same temperature. A contains $1\ \mathrm{g}$ of hydrogen and B contains $1\ \mathrm{g}$ of oxygen. $\mathrm{P_A}$ and $\mathrm{P_B}$ are the pressures of the gases in A and B respectively, then $\frac{P_A}{P_B}$ is:

(A) 4 (B) 32 (C) 8 (D) 16

Open solution

Q37. JEE Main 2024 (Online) 27th January Evening Shift

The total kinetic energy of 1 mole of oxygen at $27^{\circ}\mathrm{C}$ is : [Use universal gas constant $(R) = 8.31\ \mathrm{J/}$ mole K]

(A) 6232.5 J (B) 5670.5 J (C) 6845.5 J (D) 5942.0 J

Open solution

Q38. JEE Main 2024 (Online) 27th January Morning Shift

The average kinetic energy of a monatomic molecule is $0.414\ \mathrm{eV}$ at temperature :

(Use $K_B = 1.38 \times 10^{-23}\ \mathrm{J/mol - K}$)

(A) 3000 K (B) 3200 K (C) 1600 K (D) 1500 K

Open solution

Q39. JEE Main 2023 (Online) 15th April Morning Shift

A flask contains Hydrogen and Argon in the ratio $2:1$ by mass. The temperature of the mixture is $30^{\circ}\mathrm{C}$. The ratio of average kinetic energy per molecule of the two gases (K argon/K hydrogen) is :

(Given: Atomic Weight of $\mathrm{Ar} = 39.9$)

(A) $\frac{39.9}{2}$ (B) 2 (C) 39.9 (D) 1

Open solution

Q40. JEE Main 2023 (Online) 13th April Evening Shift

The mean free path of molecules of a certain gas at STP is $1500\ \mathrm{d}$, where d is the diameter of the gas molecules. While maintaining the standard pressure, the mean free path of the molecules at $373\ \mathrm{K}$ is approximately:

(A) $750\ \mathrm{d}$ (B) $1500\ \mathrm{d}$ (C) $2049\ \mathrm{d}$ (D) $1098\ \mathrm{d}$

Open solution

Q41. JEE Main 2023 (Online) 13th April Morning Shift

The rms speed of oxygen molecule in a vessel at particular temperature is $\left(1+\frac{5}{x}\right)^{\frac{1}{2}} v$, where v is the average speed of the molecule. The value of x will be:

(Take $\pi = \frac{22}{7}$)

(A) 4 (B) 8 (C) 28 (D) 27

Open solution

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Q42. JEE Main 2023 (Online) 12th April Morning Shift

If the r. m.s speed of chlorine molecule is $490\ \mathrm{m/s}$ at $27^{\circ}\mathrm{C}$, the r. m. s speed of argon molecules at the same temperature will be (Atomic mass of argon $= 39.9\mathrm{u}$, molecular mass of chlorine $= 70.9\mathrm{u}$)

(A) $451.7\ \mathrm{m/s}$ (B) $751.7\ \mathrm{m/s}$ (C) $551.7\ \mathrm{m/s}$ (D) $651.7\ \mathrm{m/s}$

Open solution

Q43. JEE Main 2023 (Online) 11th April Evening Shift

The root mean square speed of molecules of nitrogen gas at $27^{\circ}\mathrm{C}$ is approximately : (Given mass of a nitrogen molecule $= 4.6 \times 10^{-26}\ \mathrm{kg}$ and take Boltzmann constant $\mathrm{k_B} = 1.4 \times 10^{-23}\mathrm{JK}^{-1}$)

(A) 91 m/s (B) 1260 m/s (C) 27.4 m/s (D) 523 m/s

Open solution

Q44. JEE Main 2023 (Online) 11th April Morning Shift

Three vessels of equal volume contain gases at the same temperature and pressure. The first vessel contains neon (monoatomic), the second contains chlorine (diatomic) and third contains uranium hexafluoride (polyatomic). Arrange these on the basis of their root mean square speed (v_{rms}) and choose the correct answer from the options given below:

(A) $\mathrm{v_{rms}}(\text{ mono }) = \mathrm{v_{rms}}(\text{ dia }) = \mathrm{v_{rms}}(\text{ poly })$

(B) $\mathrm{v_{rms}}$ (mono) $> \mathrm{v_{rms}}(\text{ dia }) > \mathrm{v_{rms}}$ (poly)

(C) $\mathrm{v_{rms}}$ (dia) $< \mathrm{v_{rms}}$ (poly) $< \mathrm{v_{rms}}$ (mono)

(D) $\mathrm{v_{rms}}$ (mono) $< \mathrm{v_{rms}}$ (dia) $< \mathrm{v_{rms}}$ (poly)

Open solution

Q45. JEE Main 2023 (Online) 10th April Evening Shift

A gas mixture consists of 2 moles of oxygen and 4 moles of neon at temperature T. Neglecting all vibrational modes, the total internal energy of the system will be,

(A) 4RT (B) 16RT (C) 8RT (D) 11RT

Open solution

Q46. JEE Main 2023 (Online) 10th April Morning Shift

Match List I with List II :

	List I		List II
(A)	3 Translational degrees of freedom	(I)	Monoatomic gases
(B)	3 Translational, 2 rotational degrees of freedom	(II)	Polyatomic gases
(C)	3 Translational, 2 rotational and 1 vibrational degrees of freedom	(III)	Rigid diatomic gases
(D)	3 Translational, 3 rotational and more than one vibrational degrees of freedom	(IV)	Nonrigid diatomic gases

Choose the correct answer from the options given below:

(A) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(B) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(C) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(D) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

Open solution

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Q47. JEE Main 2023 (Online) 8th April Evening Shift

The temperature at which the kinetic energy of oxygen molecules becomes double than its value at $27^{\circ}\mathrm{C}$ is

(A) $627^{\circ}\mathrm{C}$ (B) $927^{\circ}\mathrm{C}$ (C) $327^{\circ}\mathrm{C}$ (D) $1227^{\circ}\mathrm{C}$

Open solution

Q48. JEE Main 2023 (Online) 6th April Evening Shift

The temperature of an ideal gas is increased from $200\ \mathrm{K}$ to $800\ \mathrm{K}$. If r.m.s. speed of gas at $200\ \mathrm{K}$ is v_0. Then, r.m.s. speed of the gas at $800\ \mathrm{K}$ will be:

(A) v_0 (B) $2v_0$ (C) $4v_0$ (D) $\frac{v_0}{4}$

Open solution

Q49. JEE Main 2023 (Online) 6th April Morning Shift

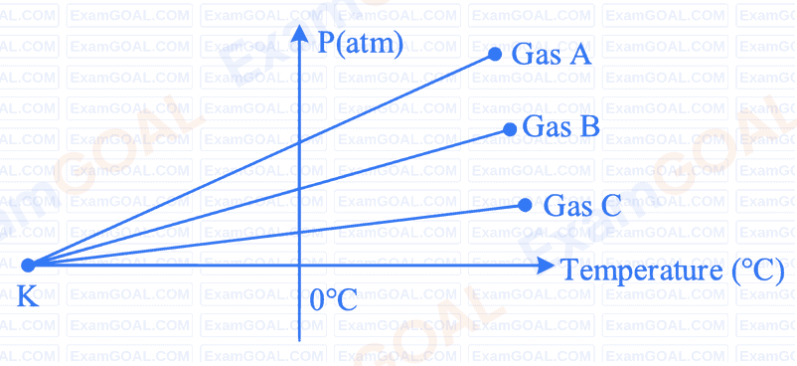
The number of air molecules per cm^3 increased from 3×10^{19} to 12×10^{19}. The ratio of collision frequency of air molecules before and after the increase in number respectively is:

(A) 1.25 (B) 0.25 (C) 0.50 (D) 0.75

Open solution

Q50. JEE Main 2023 (Online) 1st February Evening Shift

For three low density gases A, B, C pressure versus temperature graphs are plotted while keeping them at constant volume, as shown in the figure.



The temperature corresponding to the point 'K' is :

(A) $-273^{\circ}\mathrm{C}$ (B) $-373^{\circ}\mathrm{C}$ (C) $-100^{\circ}\mathrm{C}$ (D) $-40^{\circ}\mathrm{C}$

Open solution

Q51. JEE Main 2023 (Online) 1st February Morning Shift

The average kinetic energy of a molecule of the gas is

(A) proportional to volume (B) dependent on the nature of the gas

(C) proportional to absolute temperature (D) proportional to pressure

Open solution

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Q52. JEE Main 2023 (Online) 31st January Morning Shift

The correct relation between $\gamma = \frac{c_p}{c_v}$ and temperature T is :

(A) $\gamma \propto T$ (B) $\gamma \propto \frac{1}{\sqrt{T}}$

(C) $\gamma \propto \frac{1}{T}$ (D) $\gamma \propto T^0$

Open solution

Q53. JEE Main 2023 (Online) 30th January Evening Shift

A flask contains hydrogen and oxygen in the ratio of $2:1$ by mass at temperature $27^\circ\mathrm{C}$. The ratio of average kinetic energy per molecule of hydrogen and oxygen respectively is:

(A) 1 : 1 (B) 4 : 1 (C) 1 : 4 (D) 2 : 1

Open solution

Q54. JEE Main 2023 (Online) 30th January Morning Shift

The pressure (P) and temperature (T) relationship of an ideal gas obeys the equation $\mathrm{PT}^2 =$ constant. The volume expansion coefficient of the gas will be :

(A) $3T^2$ (B) $\frac{3}{T^2}$ (C) $\frac{3}{T^3}$ (D) $\frac{3}{T}$

Open solution

Q55. JEE Main 2023 (Online) 29th January Evening Shift

At 300 K, the rms speed of oxygen molecules is $\sqrt{\frac{\alpha+5}{\alpha}}$ times to that of its average speed in the gas. Then, the value of α will be

(used $\pi = \frac{22}{7}$)

(A) 27 (B) 28 (C) 24 (D) 32

Open solution

Q56. JEE Main 2023 (Online) 29th January Morning Shift

A bicycle tyre is filled with air having pressure of $270\ \mathrm{kPa}$ at $27^\circ\mathrm{C}$. The approximate pressure of the air in the tyre when the temperature increases to $36^\circ\mathrm{C}$ is

(A) 262 kPa (B) 360 kPa (C) 270 kPa (D) 278 kPa

Open solution

Q57. JEE Main 2023 (Online) 25th January Evening Shift

According to law of equipartition of energy the molar specific heat of a diatomic gas at constant volume where the molecule has one additional vibrational mode is :-

(A) $\frac{9}{2}R$ (B) $\frac{5}{2}R$ (C) $\frac{3}{2}R$ (D) $\frac{7}{2}R$

Open solution

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Q58. JEE Main 2023 (Online) 25th January Morning Shift

The root mean square velocity of molecules of gas is

(A) Proportional to temperature (T)

(B) Inversely proportional to square root of temperature $\left(\sqrt{\frac{1}{T}}\right)$

(C) Proportional to square of temperature (T^2)

(D) Proportional to square root of temperature ($\sqrt{T}$)

Open solution

Q59. JEE Main 2023 (Online) 24th January Evening Shift

Let γ_1 be the ratio of molar specific heat at constant pressure and molar specific heat at constant volume of a monoatomic gas and γ_2 be the similar ratio of diatomic gas. Considering the diatomic gas molecule as a rigid rotator, the ratio, $\frac{\gamma_1}{\gamma_2}$ is :

(A) $\frac{35}{27}$ (B) $\frac{25}{21}$ (C) $\frac{21}{25}$ (D) $\frac{27}{35}$

Open solution

Q60. JEE Main 2023 (Online) 24th January Morning Shift

Given below are two statements :

Statement I : The temperature of a gas is -73°C. When the gas is heated to 527°C, the root mean square speed of the molecules is doubled.

Statement II : The product of pressure and volume of an ideal gas will be equal to translational kinetic energy of the molecules.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Open solution

Q61. JEE Main 2022 (Online) 30th July Evening Shift

If the r.m.s speed of oxygen at room temperature is $\sqrt{56}\,\mathrm{m/s}$. The r.m.s speed of nitrogen at the same temperature is _____.

(A) $8\sqrt{2}\,\mathrm{m/s}$ (B) $8\,\mathrm{m/s}$ (C) $16\,\mathrm{m/s}$ (D) $7\,\mathrm{m/s}$

Open solution

Q62. JEE Main 2022 (Online) 29th July Evening Shift

The root mean square speed of smoke particles of mass $5 \times 10^{-17}\,\mathrm{kg}$ in their Brownian motion in air at NTP is approximately. [Given $\mathrm{k} = 1.38 \times 10^{-23}\mathrm{JK}^{-1}$]

(A) $60\,\mathrm{mm\,s^{-1}}$ (B) $12\,\mathrm{mm\,s^{-1}}$ (C) $15\,\mathrm{mm\,s^{-1}}$ (D) $36\,\mathrm{mm\,s^{-1}}$

Open solution

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Q63. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

Given below are two statements :

Statement I : The average momentum of a molecule in a sample of an ideal gas depends on temperature.

Statement II : The rms speed of oxygen molecules in a gas is v. If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become $2v$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q64. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

Which statements are correct about degrees of freedom ?

(A) A molecule with n degrees of freedom has n^2 different ways of storing energy.

(B) Each degree of freedom is associated with $\frac{1}{2}$ RT average energy per mole.

(C) A monatomic gas molecule has 1 rotational degree of freedom whereas diatomic molecule has 2 rotational degrees of freedom.

(D) CH_4 has a total of 6 degrees of freedom.

Choose the correct answer from the options given below :

(A) (B) and (C) only **(B)** (B) and (D) only **(C)** (A) and (B) only **(D)** (C) and (D) only

Q65. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

Same gas is filled in two vessels of the same volume at the same temperature. If the ratio of the number of molecules is $1 : 4$, then

A. The r.m.s. velocity of gas molecules in two vessels will be the same.

B. The ratio of pressure in these vessels will be $1 : 4$.

C. The ratio of pressure will be $1 : 1$.

D. The r.m.s. velocity of gas molecules in two vessels will be in the ratio of $1 : 4$.

Choose the correct answer from the options given below :

(A) A and C only **(B)** B and D only **(C)** A and B only **(D)** C and D only

Q66. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

A gas has n degrees of freedom. The ratio of specific heat of gas at constant volume to the specific heat of gas at constant pressure will be :

(A) $\frac{n}{n+2}$ **(B)** $\frac{n+2}{n}$ **(C)** $\frac{n}{2n+2}$ **(D)** $\frac{n}{n-2}$

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Q67. JEE Main 2022 (Online) 25th July Evening Shift

Sound travels in a mixture of two moles of helium and n moles of hydrogen. If rms speed of gas molecules in the mixture is $\sqrt{2}$ times the speed of sound, then the value of n will be :

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q68. JEE Main 2022 (Online) 25th July Morning Shift

Following statements are given :

(A) The average kinetic energy of a gas molecule decreases when the temperature is reduced.

(B) The average kinetic energy of a gas molecule increases with increase in pressure at constant temperature.

(C) The average kinetic energy of a gas molecule decreases with increase in volume.

(D) Pressure of a gas increases with increase in temperature at constant pressure.

(E) The volume of gas decreases with increase in temperature.

Choose the correct answer from the options given below :

(A) (A) and (D) only
(B) (A), (B) and (D) only
(C) (B) and (D) only
(D) (A), (B) and (E) only

Open solution

Q69. JEE Main 2022 (Online) 29th June Evening Shift

A vessel contains 16g of hydrogen and 128g of oxygen at standard temperature and pressure. The volume of the vessel in cm^3 is :

(A) 72×10^5 (B) 32×10^5 (C) 27×10^4 (D) 54×10^4

Open solution

Q70. JEE Main 2022 (Online) 28th June Evening Shift

What will be the effect on the root mean square velocity of oxygen molecules if the temperature is doubled and oxygen molecule dissociates into atomic oxygen?

(A) The velocity of atomic oxygen remains same
(B) The velocity of atomic oxygen doubles
(C) The velocity of atomic oxygen becomes half
(D) The velocity of atomic oxygen becomes four times

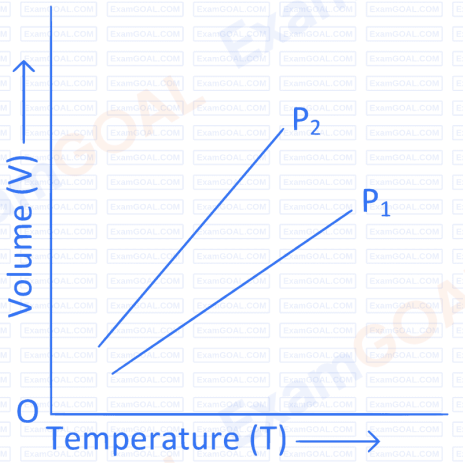
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Q71. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

For a perfect gas, two pressures P_1 and P_2 are shown in figure. The graph shows :



(A) $P_1 > P_2$

(B) $P_1 < P_2$

(C) $P_1 = P_2$

(D) Insufficient data to draw any conclusion

Q72. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

According to kinetic theory of gases,

A. The motion of the gas molecules freezes at 0°C.

B. The mean free path of gas molecules decreases if the density of molecules is increased.

C. The mean free path of gas molecules increases if temperature is increased keeping pressure constant.

D. Average kinetic energy per molecule per degree of freedom is $\frac{3}{2}k_BT$ (for monoatomic gases).

Choose the most appropriate answer from the options given below :

(A) A and C only

(B) B and C only

(C) A and B only

(D) C and D only

Q73. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

A mixture of hydrogen and oxygen has volume 2000 cm^3, temperature 300 K, pressure 100 kPa and mass 0.76 g. The ratio of number of moles of hydrogen to number of moles of oxygen in the mixture will be:

[Take gas constant R = 8.3 $JK^{-1}mol^{-1}$]

(A) $\frac{1}{3}$

(B) $\frac{3}{1}$

(C) $\frac{1}{16}$

(D) $\frac{16}{1}$

Q74. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

A flask contains argon and oxygen in the ratio of 3 : 2 in mass and the mixture is kept at 27°C. The ratio of their average kinetic energy per molecule respectively will be :

(A) 3 : 2

(B) 9 : 4

(C) 2 : 3

(D) 1 : 1

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Q75. JEE Main 2022 (Online) 25th June Evening Shift

The ratio of specific heats $\left(\frac{C_P}{C_V}\right)$ in terms of degree of freedom (f) is given by :

(A) $\left(1+\frac{f}{3}\right)$ (B) $\left(1+\frac{2}{f}\right)$ (C) $\left(1+\frac{f}{2}\right)$ (D) $\left(1+\frac{1}{f}\right)$

Open solution

Q76. JEE Main 2022 (Online) 25th June Morning Shift

The relation between root mean square speed (v_{rms}) and most probable speed (v_p) for the molar mass M of oxygen gas molecule at the temperature of 300 K will be :

(A) $v_{rms}=\sqrt{\frac{2}{3}}v_p$ (B) $v_{rms}=\sqrt{\frac{3}{2}}v_p$

(C) $v_{rms}=v_p$ (D) $v_{rms}=\sqrt{\frac{1}{3}}v_p$

Open solution

Q77. JEE Main 2021 (Online) 31st August Evening Shift

A mixture of hydrogen and oxygen has volume 500 cm^3, temperature 300 K, pressure 400 kPa and mass 0.76 g. The ratio of masses of oxygen to hydrogen will be :-

(A) 3 : 8 (B) 3 : 16 (C) 16 : 3 (D) 8 : 3

Open solution

Q78. JEE Main 2021 (Online) 27th August Evening Shift

if the rms speed of oxygen molecules at 0°C is 160 m/s, find the rms speed of hydrogen molecules at 0°C.

(A) 640 m/s (B) 40 m/s (C) 80 m/s (D) 332 m/s

Open solution

Q79. JEE Main 2021 (Online) 27th August Morning Shift

A balloon carries a total load of 185 kg at normal pressure and temperature of 27°C. What load will the balloon carry on rising to a height at which the barometric pressure is 45 cm of Hg and the temperature is −7°C. Assuming the volume constant?

(A) 181.46 kg (B) 214.15 kg (C) 219.07 kg (D) 123.54 kg

Open solution

Q80. JEE Main 2021 (Online) 27th August Morning Shift

An ideal gas is expanding such that PT^3 = constant. The coefficient of volume expansion of the gas is :

(A) $\frac{1}{T}$ (B) $\frac{2}{T}$ (C) $\frac{4}{T}$ (D) $\frac{3}{T}$

Open solution

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Q81. JEE Main 2021 (Online) 26th August Evening Shift

A cylindrical container of volume 4.0×10^{-3} m^3 contains one mole of hydrogen and two moles of carbon dioxide. Assume the temperature of the mixture is 400 K. The pressure of the mixture of gases is :

[Take gas constant as 8.3 J mol^{-1} K^{-1}]

(A) 249×10^1 Pa (B) 24.9×10^3 Pa (C) 24.9×10^5 Pa (D) 24.9 Pa

Open solution

Q82. JEE Main 2021 (Online) 26th August Morning Shift

The rms speeds of the molecules of Hydrogen, Oxygen and Carbon dioxide at the same temperature are V_H, V_O and V_C respectively then :

(A) $V_H > V_O > V_C$ (B) $V_C > V_O > V_H$ (C) $V_H = V_O > V_C$ (D) $V_H = V_O = V_C$

Open solution

Q83. JEE Main 2021 (Online) 27th July Morning Shift

The number of molecules in one litre of an ideal gas at 300 K and 2 atmospheric pressure with mean kinetic energy 2×10^{-9} J per molecules is :

(A) 0.75×10^{11} (B) 3×10^{11} (C) 1.5×10^{11} (D) 6×10^{11}

Open solution

Q84. JEE Main 2021 (Online) 22th July Evening Shift

What will be the average value of energy for a monoatomic gas in thermal equilibrium at temperature T?

(A) $\frac{3}{2}k_BT$ (B) k_BT (C) $\frac{2}{3}k_BT$ (D) $\frac{1}{2}k_BT$

Open solution

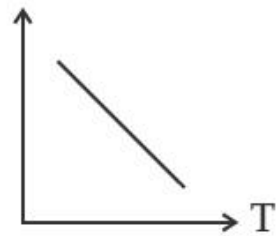
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Q85. JEE Main 2021 (Online) 20th July Evening Shift

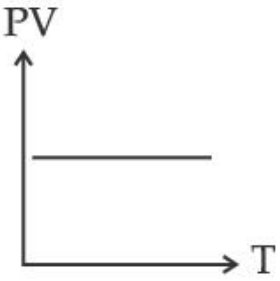
Which of the following graphs represent the behavior of an ideal gas? Symbols have their usual meaning.

Open solution

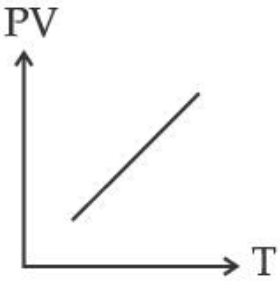
(A)



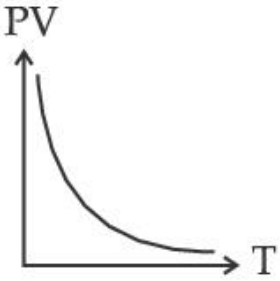
(B)



(C)



(D)



Q86. JEE Main 2021 (Online) 20th July Evening Shift

The correct relation between the degrees of freedom f and the ratio of specific heat γ is :

(A) $f = \frac{2}{\gamma - 1}$ (B) $f = \frac{2}{\gamma + 1}$ (C) $f = \frac{\gamma + 1}{2}$ (D) $f = \frac{1}{\gamma + 1}$

Open solution

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Q87. JEE Main 2021 (Online) 20th July Morning Shift

Consider a mixture of gas molecule of types A, B and C having masses $m_A < m_B < m_C$. The ratio of their root mean square speeds at normal temperature and pressure is :

(A) $v_A = v_B \neq v_C$

(B) $\frac{1}{v_A} > \frac{1}{v_B} > \frac{1}{v_C}$

(C) $\frac{1}{v_A} < \frac{1}{v_B} < \frac{1}{v_C}$

(D) $v_A = v_B = v_C = 0$

Open solution

Q88. JEE Main 2021 (Online) 18th March Evening Shift

Consider a sample of oxygen behaving like an ideal gas. At 300 K, the ratio of root mean square (rms) velocity to the average velocity of gas molecule would be :

(Molecular weight of oxygen is 32g/mol; R = 8.3 J K^{-1} mol^{-1})

(A) $\sqrt{\frac{3\pi}{8}}$

(B) $\sqrt{\frac{3}{3}}$

(C) $\sqrt{\frac{8}{3}}$

(D) $\sqrt{\frac{8\pi}{3}}$

Open solution

Q89. JEE Main 2021 (Online) 18th March Morning Shift

What will be the average value of energy along one degree of freedom for an ideal gas in thermal equilibrium at a temperature T? (k_B is Boltzmann constant)

(A) $\frac{1}{2}k_BT$

(B) $\frac{2}{3}k_BT$

(C) $\frac{3}{2}k_BT$

(D) k_BT

Open solution

Q90. JEE Main 2021 (Online) 17th March Evening Shift

If one mole of the polyatomic gas is having two vibrational modes and β is the ratio of molar specific heats for polyatomic gas $\left(\beta = \frac{C_P}{C_V}\right)$ then the value of β is :

(A) 1.02

(B) 1.35

(C) 1.2

(D) 1.25

Open solution

Q91. JEE Main 2021 (Online) 17th March Morning Shift

A polyatomic ideal gas has 24 vibrational modes. What is the value of γ?

(A) 1.37

(B) 1.30

(C) 1.03

(D) 10.3

Open solution

Q92. JEE Main 2021 (Online) 17th March Morning Shift

Two ideal polyatomic gases at temperatures T_1 and T_2 are mixed so that there is no loss of energy. If F_1 and F_2, m_1 and m_2, n_1 and n_2 be the degrees of freedom, masses, number of molecules of the first and second gas respectively, the temperature of mixture of these two gases is :

(A) $\frac{n_1F_1T_1+n_2F_2T_2}{F_1+F_2}$

(B) $\frac{n_1F_1T_1+n_2F_2T_2}{n_1F_1+n_2F_2}$

(C) $\frac{n_1T_1+n_2T_2}{n_1+n_2}$

(D) $\frac{n_1F_1T_1+n_2F_2T_2}{n_1+n_2}$

Open solution

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Q93. JEE Main 2021 (Online) 16th March Evening Shift

Calculate the value of mean free path (λ) for oxygen molecules at temperature 27°C and pressure 1.01×10^5 Pa. Assume the molecular diameter 0.3 nm and the gas is ideal. (k = 1.38×10^{-23} JK $^{-1}$)

Open solution

(A) 32 nm (B) 58 nm (C) 86 nm (D) 102 nm

Q94. JEE Main 2021 (Online) 16th March Morning Shift

The volume V of an enclosure contains a mixture of three gases, 16 g of oxygen, 28 g of nitrogen and 44 g of carbon dioxide at absolute temperature T. Consider R as universal gas constant. The pressure of the mixture of gases is :

Open solution

(A) $\frac{3RT}{V}$ (B) $\frac{4RT}{V}$ (C) $\frac{88RT}{V}$ (D) $\frac{5}{2}\frac{RT}{V}$

Q95. JEE Main 2021 (Online) 26th February Evening Shift

The internal energy (U), pressure (P) and volume (V) of an ideal gas are related as U = 3PV + 4. The gas is :

Open solution

(A) either monoatomic or diatomic. (B) monoatomic only.

(C) polyatomic only. (D) diatomic only.

Q96. JEE Main 2021 (Online) 25th February Evening Shift

Given below are two statements :

Open solution

Statement I : In a diatomic molecule, the rotational energy at a given temperature obeys Maxwell's distribution.

Statement II : In a diatomic molecule, the rotational energy at a given temperature equals the translational kinetic energy for each molecule.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

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Q97. JEE Main 2021 (Online) 24th February Evening Shift

On the basis of kinetic theory of gases, the gas exerts pressure because its molecules :

(A) continuously lose their energy till it reaches wall.

(B) are attracted by the walls of container.

(C) suffer change in momentum when impinge on the walls of container.

(D) continuously stick to the walls of container.

Open solution

Q98. JEE Main 2020 (Online) 6th September Evening Slot

In a dilute gas at pressure P and temperature T, the mean time between successive collisions of a molecule varies with T as :

(A) $\sqrt{T}$ (B) T (C) $\frac{1}{T}$ (D) $\frac{1}{\sqrt{T}}$

Open solution

Q99. JEE Main 2020 (Online) 6th September Morning Slot

Molecules of an ideal gas are known to have three translational degrees of freedom and two rotational degrees of freedom.The gas is maintained at a temperature of T. The total internal energy, U of a mole of this gas, and the value of $\gamma \left(= \frac{C_p}{C_v}\right)$ are given, respectively by:

(A) U = $\frac{5}{2}RT$ and $\gamma = \frac{7}{5}$ (B) U = 5RT and $\gamma = \frac{6}{5}$

(C) U = 5RT and $\gamma = \frac{7}{5}$ (D) U = $\frac{5}{2}RT$ and $\gamma = \frac{6}{5}$

Open solution

Q100. JEE Main 2020 (Online) 5th September Morning Slot

Number of molecules in a volume of 4 cm^3 of a perfect monoatomic gas at some temperature T and at a pressure of 2 cm of mercury is close to?
(Given, mean kinetic energy of a molecule (at T) is 4×10^{-14} erg, g = 980 cm/s^2, density of mercury = 13.6 g/cm^3)

(A) 5.8×10^{18} (B) 4.0×10^{16} (C) 5.8×10^{16} (D) 4.0×10^{18}

Open solution

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Q101. JEE Main 2020 (Online) 4th September Morning Slot

Match the $\frac{C_P}{C_V}$ ratio for ideal gases with different type of molecules :

Open solution

Molecule Type	C_P/C_V
(A) Monatomic	(I) 7/5
(B) Diatomic rigid molecules	(II) 9/7
(C) Diatomic non-rigid molecules	(III) 4/3
(D) Triatomic rigid molecules	(IV) 5/3

(A) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(B) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)
(C) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
(D) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Q102. JEE Main 2020 (Online) 3rd September Evening Slot

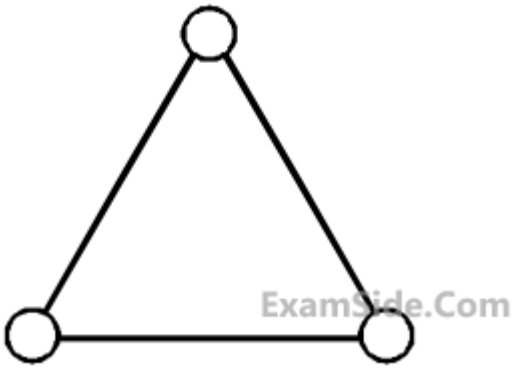
Open solution

To raise the temperature of a certain mass of gas by 50°C at a constant pressure, 160 calories of heat is required. When the same mass of gas is cooled by 100°C at constant volume, 240 calories of heat is released. How many degrees of freedom does each molecule of this gas have (assume gas to be ideal)?

(A) 6 **(B)** 7 **(C)** 5 **(D)** 3

Q103. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution



Consider a gas of triatomic molecules. The molecules are assumed to be triangular and made of massless rigid rods whose vertices are occupied by atoms. The internal energy of a mole of the gas at temperature T is :

(A) $\frac{3}{2}RT$ **(B)** $\frac{9}{2}RT$ **(C)** $\frac{5}{2}RT$ **(D)** 3RT

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Q104. JEE Main 2020 (Online) 2nd September Evening Slot

An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statements are true?
(A) the mean free path of the molecules decreases.
(B) the mean collision time between the molecules decreases.
(C) the mean free path remains unchanged.
(D) the mean collision time remains unchanged.

(A) (C) and (D) **(B)** (A) and (D) **(C)** (B) and (C) **(D)** (A) and (B)

Open solution

Q105. JEE Main 2020 (Online) 2nd September Morning Slot

A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. Assuming the gases to be ideal and the oxygen bond to be rigid, the total internal energy (in units of RT) of the mixture is :

(A) 11 **(B)** 20 **(C)** 15 **(D)** 13

Open solution

Q106. JEE Main 2020 (Online) 9th January Evening Slot

Two gases-argon (atomic radius 0.07 nm, atomic weight 40) and xenon (atomic radius 0.1 nm, atomic weight 140) have the same number density and are at the same temperature. The ratio of their respective mean free times is closest to :

(A) 2.3 **(B)** 1.83 **(C)** 4.67 **(D)** 3.67

Open solution

Q107. JEE Main 2020 (Online) 9th January Morning Slot

Consider two ideal diatomic gases A and B at some temperature T. Molecules of the gas A are rigid, and have a mass m. Molecules of the gas B have an additional vibrational mode, and have a mass $\frac{m}{4}$. The ratio of the specific heats (C_V^A and C_V^B) of gas A and B, respectively is :

(A) 7 : 9 **(B)** 5 : 7 **(C)** 3 : 5 **(D)** 5 : 9

Open solution

Q108. JEE Main 2020 (Online) 8th January Evening Slot

Consider a mixture of n moles of helium gas and 2n moles of oxygen gas (molecules taken to be rigid) as an ideal gas. Its C_P/C_V value will be :

(A) 23/15 **(B)** 67/45 **(C)** 40/27 **(D)** 19/13

Open solution

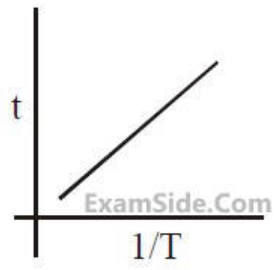
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Q109. JEE Main 2020 (Online) 8th January Morning Slot

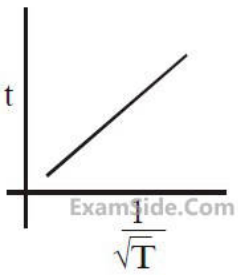
Open solution

The plot that depicts the behavior of the mean free time t (time between two successive collisions) for the molecules of an ideal gas, as a function of temperature (T), qualitatively, is:
(Graphs are schematic and not drawn to scale)

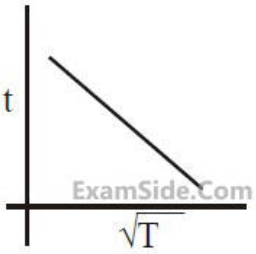
(A)



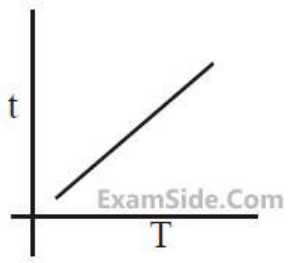
(B)



(C)



(D)



Q110. JEE Main 2020 (Online) 7th January Evening Slot

Open solution

Under an adiabatic process, the volume of an ideal gas gets doubled. Consequently the mean collision time between the gas molecule changes from τ_1 to τ_2 . If $\frac{C_p}{C_v} = \gamma$ for this gas then a good estimate for $\frac{\tau_2}{\tau_1}$ is given by :

(A) $(2)^{\frac{1+\gamma}{2}}$

(B) 2

(C) $\left(\frac{1}{2}\right)^{\frac{1+\gamma}{2}}$

(D) $\left(\frac{1}{2}\right)^{\gamma}$

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Q111. JEE Main 2020 (Online) 7th January Morning Slot

Two moles of an ideal gas with $\frac{C_P}{C_V} = \frac{5}{3}$ are mixed with 3 moles of another ideal gas with $\frac{C_P}{C_V} = \frac{4}{3}$. The value of $\frac{C_P}{C_V}$ for the mixture is :

(A) 1.50 (B) 1.45 (C) 1.47 (D) 1.42

Open solution

Q112. JEE Main 2019 (Online) 12th April Evening Slot

The number density of molecules of a gas depends on their distance r from the origin as, $n(r) = n_0 e^{-\alpha r^4}$. Then the total number of molecules is proportional to :

(A) $n_0\alpha^{-3/4}$ (B) $n_0\alpha^{-3}$ (C) $n_0\alpha^{1/4}$ (D) $\sqrt{n_0}\alpha^{1/2}$

Open solution

Q113. JEE Main 2019 (Online) 12th April Morning Slot

Two moles of helium gas is mixed with three moles of hydrogen molecules (taken to be rigid). What is the molar specific heat of mixture at constant volume ? (R = 8.3 J/mol K)

(A) 21.6 J/mol K (B) 17.4 J/mol K (C) 15.7 J/mol K (D) 19.7 J/mol K

Open solution

Q114. JEE Main 2019 (Online) 10th April Morning Slot

A 25×10^{-3} m^3 volume cylinder is filled with 1 mol of O_2 gas at room temperature (300K). The molecular diameter of O_2, and its root mean square speed, are found to be 0.3 nm, and 200 m/s, respectively. What is the average collision rate (per second) for an O_2 molecule ?

(A) $\sim 10^{13}$ (B) $\sim 10^{12}$ (C) $\sim 10^{11}$ (D) $\sim 10^{10}$

Open solution

Q115. JEE Main 2019 (Online) 9th April Evening Slot

The specific heats, C_P and C_V of a gas of diatomic molecules, A, are given (in units of J mol^{-1} K^{-1}) by 29 and 22, respectively. Another gas of diatomic molecules, B, has the corresponding values 30 and 21. If they are treated as ideal gases, then :-

(A) A is rigid but B has a vibrational mode

(B) A has a vibrational mode but B has none

(C) A has one vibrational mode and B has two

(D) Both A and B have a vibrational mode each

Open solution

Q116. JEE Main 2019 (Online) 9th April Morning Slot

An HCl molecule has rotational, translational and vibrational motions. If the rms velocity of HCl molecules in its gaseous phase is $\overline{v}$, m is its mass and k_B is Boltzmann constant, then its temperature will be :

(A) $\frac{m\overline{v}^2}{5k_B}$ (B) $\frac{m\overline{v}^2}{6k_B}$

(C) $\frac{m\overline{v}^2}{7k_B}$ (D) $\frac{m\overline{v}^2}{3k_B}$

Open solution

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Q117. JEE Main 2019 (Online) 9th April Morning Slot

For a given gas at 1 atm pressure, rms speed of the molecule is 200 m/s at 127°C. At 2 atm pressure and at 227°C, the rms speed of the molecules will be :

(A) 100 m/s (B) $100\sqrt{5}$ m/s (C) $80\sqrt{5}$ m/s (D) 80 m/s

Open solution

Q118. JEE Main 2019 (Online) 8th April Evening Slot

The temperature, at which the root mean square velocity of hydrogen molecules equals their escape velocity from the earth, is closest to :

[Boltzmann Constant $k_B = 1.38 \times 10^{-23}$ J/K Avogadro Number $N_A = 6.02 \times 10^{26}$ /kg Radius of Earth : 6.4×10^6 m Gravitational acceleration on Earth = $10ms^{-2}$]

(A) 3×10^5 K (B) 10^4 K (C) 650 K (D) 800 K

Open solution

Q119. JEE Main 2019 (Online) 8th April Morning Slot

If 10^{22} gas molecules each of mass 10^{-26} kg collide with a surface (perpendicular to it) elastically per second over an area 1 m^2 with a speed 10^4 m/s, the pressure exerted by the gas molecules will be of the order of :

(A) 10^8 N/m^2 (B) 10^{16} N/m^2 (C) 10^4 N/m^2 (D) 2 N/m^2

Open solution

Q120. JEE Main 2019 (Online) 12th January Evening Slot

An ideal gas is enclosed in a cylinder at pressure of 2 atm and temperature 300 K. The mean time between two successive collisions is 6×10^{-8} s. If the pressure is doubled and temperature is increased to 500 K, the mean time between two successive collisions will be close to

(A) 0.5×10^{-8} s (B) 4×10^{-8} s (C) 3×10^{-6} s (D) 2×10^{-7} s

Open solution

Q121. JEE Main 2019 (Online) 12th January Evening Slot

A vertical closed cylinder is separated into two parts by a frictionless piston of mass m and of negligible thickness. The piston is free to move along the length of the cylinder. The length of the cylinder above the piston is ℓ_1, and that below the piston is ℓ_2, such that $\ell_1 > \ell_2$. Each part of the cylinder contains n moles of an ideal gas at equal temperature T. If the piston is stationary, its mass, m, will be given by :

(R is universal gas constant and g is the acceleration due to gravity)

(A) $\frac{nRT}{g}\left[\frac{\ell_1-\ell_2}{\ell_1\ell_2}\right]$

(B) $\frac{RT}{g}\left[\frac{2\ell_1+\ell_2}{\ell_1\ell_2}\right]$

(C) $\frac{nRT}{g}\left[\frac{1}{\ell_2}+\frac{1}{\ell_1}\right]$

(D) $\frac{RT}{ng}\left[\frac{\ell_1-3\ell_2}{\ell_1\ell_2}\right]$

Open solution

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Q122. JEE Main 2019 (Online) 12th January Morning Slot

An ideal gas occupies a volume of $2m^3$ at a pressure of 3×10^6 Pa. The energy of the gas is :

(A) 6×10^4 J (B) 9×10^6 J (C) 3×10^2 J (D) 10^8 J

Open solution

Q123. JEE Main 2019 (Online) 11th January Morning Slot

A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. considering only translational and rotational modes, the total internal energy of the system is :

(A) 12 RT (B) 20 RT (C) 4 RT (D) 15 RT

Open solution

Q124. JEE Main 2019 (Online) 10th January Evening Slot

Two kg of a monoatomic gas is at a pressure of 4×10^4 N/m^2. The density of the gas is 8 kg/m^3. What is the order of energy of the gas due to its thermal motion ?

(A) 10^4 J (B) 10^3 J (C) 10^5 J (D) 10^6 J

Open solution

Q125. JEE Main 2019 (Online) 9th January Morning Slot

A mixture of 2 moles of helium gas (atomic mass = 4 u), and 1 mole of argon gas (atomic mass = 40 u) is kept at 300 K in a container. The ratio of their rms speeds $\left[\frac{V_{rms}\,(helium)}{V_{rms}\,(\arg on)}\right]$, is close to :

(A) 3.16 (B) 0.32 (C) 0.45 (D) 2.24

Open solution

Q126. JEE Main 2018 (Online) 16th April Morning Slot

Two moles of helium are mixed with n moles of hydrogen. If $\frac{Cp}{Cv} = \frac{3}{2}$ for the mixture, then the value of n is :

(A) 1 (B) 3 (C) 2 (D) 3 / 2

Open solution

Q127. JEE Main 2018 (Offline)

The mass of a hydrogen molecule is 3.32×10^{-27} kg. If 10^{23} hydrogen molecules strike, per second, a fixed wall of area 2 cm^2 at an angle of 45° to the normal, and rebound elastically with a speed of 10^3 m/s, then the pressure on the wall is nearly:

(A) 2.35×10^3 N m^{-2} (B) 4.70×10^3 N m^{-2}

(C) 2.35×10^2 N m^{-2} (D) 4.70×10^2 N m^{-2}

Open solution

Q128. JEE Main 2018 (Online) 15th April Evening Slot

The value closest to the thermal velocity of a Helium atom at room temperature (300 K) in ms^{-1} is : [$k_B = 1.4 \times 10^{-23}$ J/K; $m_{He} = 7 \times 10^{-27}$ kg]

(A) 1.3×10^4 (B) 1.3×10^3 (C) 1.3×10^5 (D) 1.3×10^2

Open solution

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Q129. JEE Main 2017 (Online) 9th April Morning Slot

N moles of a diatomic gas in a cylinder are at a temperature T. Heat is supplied to the cylinder such that the temperature remains constant but n moles of the diatomic gas get converted into monoatomic gas. What is the change in the total kinetic energy of the gas ?

Open solution

(A) $\frac{1}{2}$ nRT (B) 0 (C) $\frac{3}{2}$ nRT (D) $\frac{5}{2}$ nRT

Q130. JEE Main 2017 (Online) 8th April Morning Slot

An ideal gas has molecules with 5 degrees of freedom. The ratio of specific heats at constant pressure (C_p) and at constant volume (C_v) is :

Open solution

(A) 6 (B) $\frac{7}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{5}$

Q131. JEE Main 2017 (Offline)

The temperature of an open room of volume 30 m^3 increases from 17°C to 27°C due to the sunshine. The atmospheric pressure in the room remains 1×10^5 Pa. If N_i and N_f are the number of molecules in the room before and after heating, then $N_f - N_i$ will be :

Open solution

(A) -1.61×10^{23} (B) 1.38×10^{23} (C) 2.5×10^{25} (D) -2.5×10^{25}

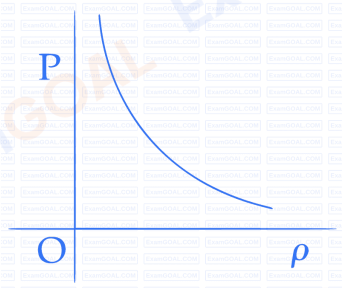
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Q132. JEE Main 2016 (Online) 10th April Morning Slot

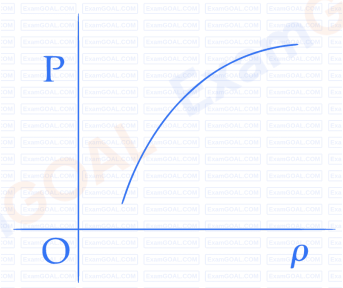
Which of the following shows the correct relationship between the pressure 'P' and density ρ of an ideal gas at constant temperature ?

Open solution

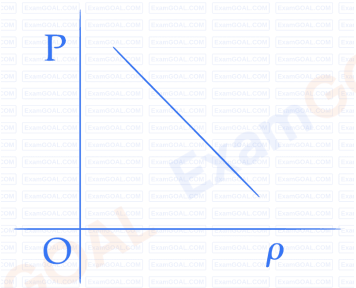
(A)



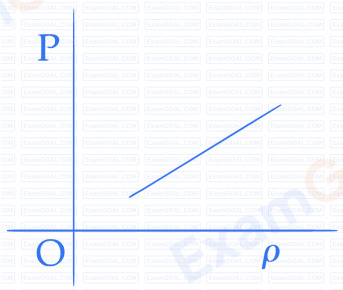
(B)



(C)



(D)



Q133. JEE Main 2015 (Online) 11th April Morning Slot

Using equipartition of energy, the specific heat (in J kg $^{-1}$ K^{-1}) of aluminium at room temperature can be estimated to be (atomic weight of aluminium = 27)

Open solution

(A) 25 (B) 410 (C) 925 (D) 1850

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Q134. JEE Main 2015 (Online) 10th April Morning Slot

In an ideal gas at temperature T, the average force that a molecule applies on the walls of a closed container depends on T as T^q . A good estimate for q is :

(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Open solution

Q135. JEE Main 2015 (Offline)

Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as V^q, where V is the volume of the gas. The value of q is: $\left(\gamma = \frac{C_p}{C_v}\right)$

(A) $\frac{\gamma+1}{2}$ (B) $\frac{\gamma-1}{2}$ (C) $\frac{3\gamma+5}{6}$ (D) $\frac{3\gamma-5}{6}$

Open solution

Q136. JEE Main 2014 (Online) 19th April Morning Slot

A gas molecule of mass M at the surface of the Earth has kinetic energy equivalent to 0^oC. If it were to go up straight without colliding with any other molecules, how high it would rise ? Assume that the height attained is much less than radius of the earth. (k_B is Boltzmann constant)

(A) 0 (B) $\frac{273k_B}{2Mg}$ (C) $\frac{546k_B}{3Mg}$ (D) $\frac{819k_B}{2Mg}$

Open solution

Q137. JEE Main 2014 (Online) 12th April Morning Slot

At room temperature a diatomic gas is found to have an r.m.s. speed of 1930 ms^{-1}. The gas is :

(A) H_2 (B) Cl_2 (C) O_2 (D) F_2

Open solution

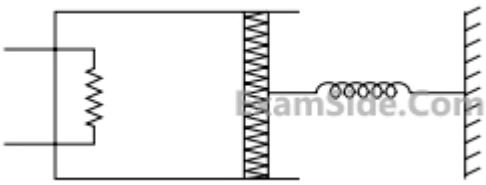
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Q138. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

An ideal monoatomic gas is confined in a cylinder by a spring loaded piston of cross section 8.0×10^{-3} m^2. Initially the gas is at 300K and occupies a volume of 2.4×10^{-3} m^3 and the spring is in its relaxed state as shown in figure. The gas is heated by a small heater until the piston moves out slowly by 0.1 m. The force constant of the spring is 8000 N/m and the atmospheric pressure is 1.0×10^5 N/m^2. The cylinder and the piston are thermally insulated. The piston and the spring are massless and there is no friction between the piston and the cylinder. The final temperature of the gas will be :

(Neglect the heat loss through the lead wires of the heater. The heat capacity of the heater coil is also negligible)



(A) 300 K (B) 800 K (C) 500 K (D) 1000 K

Q139. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

Modern vacuum pumps can evacuate a vessel down to a pressure of 4.0×10^{-15} atm. at room temperature (300 K). Taking R = 8.3 JK^{-1} $mole^{-1}$, 1 atm = 10^5 Pa and $N_{Avogadro} = 6 \times 10^{23}$ $mole^{-1}$, the mean distance between molecules of gas in an evacuated vessel will be of the order of :

(A) 0.2 μm (B) 0.2 mm (C) 0.2 cm (D) 0.2 nm

Q140. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

In the isothermal expansion of 10 g of gas from volume V to 2V the work done by the gas is 575 J. What is the root mean square speed of the molecules of the gas at that temperature?

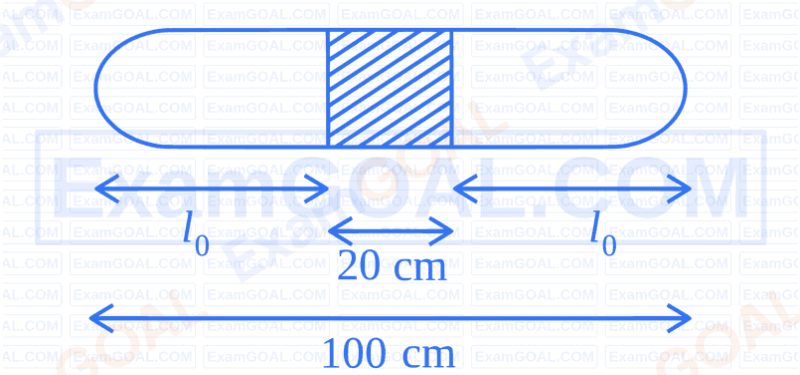
(A) 398 m/s (B) 520 m/s (C) 499 m/s (D) 532 m/s

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Q141. JEE Main 2013 (Online) 23th April Morning Slot

A thin tube sealed at both ends is 100 cm long. It lies horizontally, the middle 20 cm containing mercury and two equal ends containing air at standard atmospheric pressure. If the tube is now turned to a vertical position, by what amount will the mercury be displaced?
(Given : cross-section of the tube can be assumed to be uniform)

Open solution

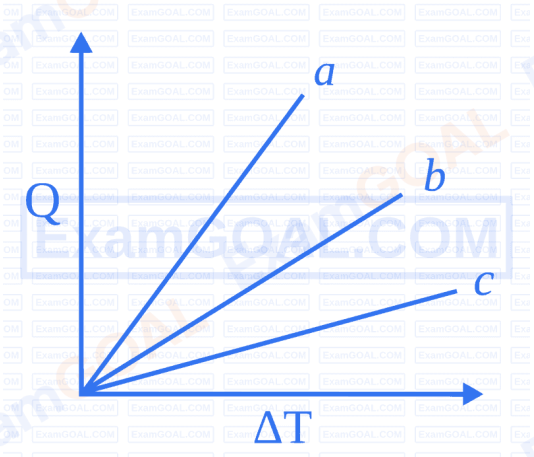


(A) 2.95 cm (B) 5.18 cm (C) 8.65 cm (D) 0.0 cm

Q142. JEE Main 2013 (Online) 9th April Morning Slot

Figure shows the variation in temperature (ΔT) with the amount of heat supplied (Q) in an isobaric process corresponding to a monoatomic (M), diatomic (D) and a polyatomic (P) gas. The initial state of all the gases are the same and the scales for the two axes coincide. Ignoring vibrational degrees of freedom, the lines a, b and c respectively correspond to :

Open solution



(A) P, M and D (B) M, D and P (C) P, D and M (D) D, M and P

Q143. AIEEE 2012 (Online) 7th May

A perfect gas at $27^{\circ}\mathrm{C}$ is heated at constant pressure so as to double its volume. The final temperature of the gas will be, close to

Open solution

(A) $327^{\circ}\mathrm{C}$ (B) $200^{\circ}\mathrm{C}$ (C) $54^{\circ}\mathrm{C}$ (D) $300^{\circ}\mathrm{C}$

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Q144. AIEEE 2011

Three perfect gases at absolute temperatures T_1, T_2 and T_3 are mixed. The masses of molecules are m_1, m_2 and m_3 and the number of molecules are n_1, n_2 and n_3 respectively. Assuming no loss of energy, the final temperature of the mixture is:

(A) $\frac{n_1T_1+n_2T_2+n_3T_3}{n_1+n_2+n_3}$

(B) $\frac{n_1T_1^2+n_2T_2^2+n_3T_3^2}{n_1T_1+n_2T_2+n_3T_3}$

(C) $\frac{n_1^2T_1^2+n_2^2T_2^2+n_3^2T_3^2}{n_1T_1+n_2T_2+n_3T_3}$

(D) $\frac{(T_1+T_2+T_3)}{3}$

Open solution

Q145. AIEEE 2009

One kg of a diatomic gas is at a pressure of $8 \times 10^4\ N/m^2$. The density of the gas is $4kg/m^3$. What is the energy of the gas due to its thermal motion ?

(A) $5 \times 10^4\ J$ (B) $6 \times 10^4\ J$ (C) $7 \times 10^4\ J$ (D) $3 \times 10^4\ J$

Open solution

Q146. AIEEE 2006

Two rigid boxes containing different ideal gases are placed on a table. Box A contains one mole of nitrogen at temperature T_0, while Box contains one mole of helium at temperature $\left(\frac{7}{3}\right)T_0$. The boxes are then put into thermal contact with each other, and heat flows between them until the gases reach a common final temperature (ignore the heat capacity of boxes). Then, the final temperature of the gases, T_f in terms of T_0 is

(A) $T_f = \frac{3}{7}T_0$ (B) $T_f = \frac{7}{3}T_0$ (C) $T_f = \frac{3}{2}T_0$ (D) $T_f = \frac{5}{2}T_0$

Open solution

Q147. AIEEE 2005

A gaseous mixture consists of $16\ g$ of helium and $16\ g$ of oxygen. The ratio $\frac{C_p}{C_v}$ of the mixture is

(A) 1.62 (B) 1.59 (C) 1.54 (D) 1.4

Open solution

Q148. AIEEE 2004

One mole of ideal monatomic gas $(\gamma = 5/3)$ is mixed with one mole of diatomic gas $(\gamma = 7/5)$. What is γ for the mixture? γ Denotes the ratio of specific heat at constant pressure, to that at constant volume

(A) $35/23$ (B) $23/15$ (C) $3/2$ (D) $4/3$

Open solution

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Q149. AIEEE 2002

Cooking gas containers are kept in a lorry moving with uniform speed. The temperature of the gas molecules inside will

(A) increase

(B) decrease

(C) remain same

(D) decrease for some, while increase for others

Open solution

Q150. AIEEE 2002

At what temperature is the $r.m.s$ velocity of a hydrogen molecule equal to that of an oxygen molecule at $47^{\circ}C$?

(A) $80K$ (B) $-73\ K$ (C) $3\ K$ (D) $20\ K$

Open solution

Q151. AIEEE 2002

1 mole of a gas with $\gamma = 7/5$ is mixed with 1 mole of a gas with $\gamma = 5/3$, then the value of γ for the resulting mixture is

(A) $7/5$ (B) $2/5$ (C) $3/2$ (D) $12/7$

Open solution

Numerical 16 questions

Q152. JEE Main 2026 (Online) 24th January Morning Shift

A gas of certain mass filled in a closed cylinder at a pressure of 3.23 kPa has temperature $50^{\circ}\mathrm{C}$. The gas is now heated to double its temperature. The modified pressure is ____ Pa .

Open solution

Q153. JEE Main 2026 (Online) 22nd January Evening Shift

An insulated cylinder of volume $60\ \mathrm{cm}^3$ is filled with a gas at $27^{\circ}\mathrm{C}$ and 2 atmospheric pressure. Then the gas is compressed making the final volume as $20\ \mathrm{cm}^3$ while allowing the temperature to rise to $77^{\circ}\mathrm{C}$. The final pressure is ____ atmospheric pressure.

Open solution

Q154. JEE Main 2025 (Online) 2nd April Evening Shift

The internal energy of air in $4\ \mathrm{m} \times 4\ \mathrm{m} \times 3\ \mathrm{m}$ sized room at 1 atmospheric pressure will be_____$\times 10^6\ \mathrm{J}$

(Consider air as diatomic molecule)

Open solution

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Q155. JEE Main 2025 (Online) 2nd April Morning Shift

γ_A is the specific heat ratio of monoatomic gas A having 3 translational degrees of freedom. γ_B is the specific heat ratio of polyatomic gas B having 3 translational, 3 rotational degrees of freedom and 1 vibrational mode. If $\frac{\gamma_A}{\gamma_B} = \left(1 + \frac{1}{n}\right)$, then the value of n is _____ .

Open solution

Q156. JEE Main 2025 (Online) 29th January Morning Shift

A container of fixed volume contains a gas at 27°C. To double the pressure of the gas, the temperature of gas should be raised to ____ °C.

Open solution

Q157. JEE Main 2022 (Online) 29th July Morning Shift

One mole of a monoatomic gas is mixed with three moles of a diatomic gas. The molecular specific heat of mixture at constant volume is $\frac{\alpha^2}{4}\mathrm{R\,J/mol\,K}$; then the value of α will be _____. (Assume that the given diatomic gas has no vibrational mode).

Open solution

Q158. JEE Main 2022 (Online) 28th June Morning Shift

The total internal energy of two mole monoatomic ideal gas at temperature T = 300 K will be _____ J. (Given R = 8.31 J/mol.K)

Open solution

Q159. JEE Main 2022 (Online) 25th June Evening Shift

When a gas filled in a closed vessel is heated by raising the temperature by 1°C, its pressure increases by 0.4%. The initial temperature of the gas is _____ K.

Open solution

Q160. JEE Main 2021 (Online) 1st September Evening Shift

The average translational kinetic energy of N_2 gas molecules at .....°C becomes equal to the K.E. of an electron accelerated from rest through a potential difference of 0.1 volt. (Given k_B = 1.38 $\times$ 10^{-23} J/K) (Fill the nearest integer).

Open solution

Q161. JEE Main 2021 (Online) 25th July Evening Shift

A system consists of two types of gas molecules A and B having same number density 2 $\times$ $10^{25}/m^3$. The diameter of A and B are 10 $\overset{o}{A}$ and 5 $\overset{o}{A}$ respectively. They suffer collision at room temperature. The ratio of average distance covered by the molecule A to that of B between two successive collision is _____ $\times$ 10^{-2}

Open solution

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Q162. JEE Main 2021 (Online) 26th February Morning Shift

A container is divided into two chambers by a partition. The volume of first chamber is 4.5 litre and second chamber is 5.5 litre. The first chamber contain 3.0 moles of gas at pressure 2.0 atm and second chamber contain 4.0 moles of gas at pressure 3.0 atm. After the partition is removed and the mixture attains equilibrium, then, the common equilibrium pressure existing in the mixture is x $\times$ 10^{-1} atm. Value of x is _____.

Open solution

Q163. JEE Main 2021 (Online) 24th February Evening Shift

The root mean square speed of molecules of a given mass of a gas at 27°C and 1 atmosphere pressure is 200 ms^{-1}. The root mean square speed of molecules of the gas at 127°C and 2 atmosphere pressure is $\frac{x}{\sqrt{3}}$ ms^{-1}. The value of x will be _____.

Open solution

Q164. JEE Main 2020 (Online) 6th September Morning Slot

Initially a gas of diatomic molecules is contained in a cylinder of volume V_1 at a pressure P_1 and temperature 250 K. Assuming that 25% of the molecules get dissociated causing a change in number of moles. The pressure of the resulting gas at temperature 2000 K, when contained in a volume $2V_1$ is given by P_2 . The ratio $\frac{P_2}{P_1}$ is _____.

Open solution

Q165. JEE Main 2020 (Online) 5th September Evening Slot

Nitrogen gas is at 300°C temperature. The temperature (in K) at which the rms speed of a H_2 molecule would be equal to the rms speed of a nitrogen molecule, is _____.
(Molar mass of N_2 gas 28 g).

Open solution

Q166. JEE Main 2020 (Online) 4th September Evening Slot

The change in the magnitude of the volume of an ideal gas when a small additional pressure ΔP is applied at a constant temperature, is the same as the change when the temperature is reduced by a small quantity ΔT at constant pressure. The initial temperature and pressure of the gas were 300 K and 2 atm. respectively.
If $|\Delta T| = C|\Delta P|$ then value of C in (K/atm.) is _____.

Open solution

Q167. JEE Main 2020 (Online) 4th September Morning Slot

A closed vessel contains 0.1 mole of a monoatomic ideal gas at 200 K. If 0.05 mole of the same gas at 400 K is added to it, the final equilibrium temperature (in K) of the gas in the vessel will be close to _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **A**	3. **B**	4. **C**	5. **B**
6. **C**	7. **C**	8. *Scan QR*	9. **B**	10. **D**
11. **A**	12. **D**	13. **D**	14. **D**	15. **A**
16. **D**	17. **C**	18. **A**	19. **D**	20. **B**
21. **B**	22. **D**	23. **C**	24. **A**	25. **B**
26. **B**	27. **A**	28. **D**	29. **D**	30. **B**
31. **A**	32. **B**	33. **A**	34. **A**	35. **C**
36. **D**	37. **A**	38. **B**	39. **D**	40. **C**
41. **C**	42. **D**	43. **D**	44. **B**	45. **D**
46. **A**	47. **C**	48. **B**	49. **B**	50. **A**
51. **C**	52. **D**	53. **A**	54. **D**	55. **B**
56. **D**	57. **D**	58. **D**	59. **B**	60. **A**
61. **B**	62. **C**	63. **D**	64. **B**	65. **C**
66. **A**	67. **B**	68. *Scan QR*	69. **C**	70. **B**
71. **A**	72. **B**	73. **B**	74. **D**	75. **B**
76. **B**	77. **C**	78. **A**	79. **D**	80. **C**
81. **C**	82. **A**	83. **C**	84. **A**	85. **C**
86. **A**	87. **C**	88. **A**	89. **A**	90. **C**
91. **C**	92. **B**	93. **D**	94. **D**	95. **C**
96. **C**	97. **C**	98. **D**	99. **A**	100. **D**
101. **C**	102. **A**	103. **D**	104. **C**	105. **C**
106. **B**	107. **B**	108. **D**	109. **B**	110. **A**
111. **D**	112. **A**	113. **B**	114. **D**	115. **B**
116. **D**	117. **B**	118. **B**	119. **D**	120. **B**
121. **A**	122. **B**	123. **D**	124. **A**	125. **A**
126. **C**	127. **A**	128. **B**	129. **A**	130. **D**
131. **D**	132. **D**	133. **C**	134. **B**	135. **A**
136. **D**	137. **A**	138. **B**	139. **B**	140. **C**
141. **B**	142. **C**	143. **A**	144. **A**	145. **A**
146. **C**	147. **A**	148. **C**	149. **C**	150. **D**

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151. **C**

Numerical

152. **3730 or 6460**	153. **7**	154. **12**	155. **3**	156. **327**
157. **3**	158. **7479**	159. **250**	160. **500**	161. **25**
162. **25**	163. **400**	164. **5**	165. **40 to 41**	166. **150**
167. **266**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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